CHAPTER 9

Combinatorics and counting

S22, S25, S30, S35

Before you begin · Sets and choices

Without a calculator, in six minutes. Answer briefly. Use choice trees and factorials. Then choose the targeted review indicated under the questions.

1. Count the outcomes of two coin tosses.

2. Calculate $4 \times 3 \times 2 \times 1$.

3. Distinguish "choose 2 pupils" from "appoint a president then a vice-president".

4. How many choices of a vowel then a consonant, with 5 vowels and 21 consonants?

5. Write $A \cap \overline{A}$.

6. List the two-element subsets of $\{a, b, c\}$.

Answers

4; 24; the second choice is ordered; 105; $\varnothing$; $\{a,b\}, \{a,c\}, \{b,c\}$.

Targeted review after the diagnostic check

Redo only the questions you got wrong. For each question: **1.** Draw the four pairs of results; R31. **2.** Perform multiplications from left to right. **3.** Compare (A,B) and (B,A) according to the students' roles. **4.** Use two levels in the tree; R31. **5.** Look for a element that would satisfy both conditions. **6.** List without changing the order of writing to avoid duplicates. The R questions serve as follow-up material with the teacher, even if their collective routine is planned for later.

Objectives

Choose a counting model; distinguish order and repetition; use binomial coefficients and Pascal's rule.

Course notes

Add or multiply

For finite pairwise disjoint sets A_i, $|\bigcup A_i| = \sum |A_i|$. If the cases overlap, we must correct double counting: $|A \cup B| = |A| + |B| - |A \cap B|$. The number of elements of the product Cartesian $A \times B$ is $|A||B|$. For a sequence of choices, multiply the numbers of possibilities at each stage, provided these numbers are valid for each previous choice considered.

Example. A code of two letters chosen from 26, followed by three digits, with repetition allowed, can take $26^2 \times 10^3$ values. A three-digit number does not start with 0: there are 9×10^2.

Lists and subsets

In a set of n elements, there are n^k ordered lists of length k with repetition; $n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$ lists of k distinct elements, if $0 \leq k \leq n$; $n!$ permutations of all elements. We set $0! = 1$.

Combinations

A combination of k elements among n is a subset of k elements; order does not matter.

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad \binom{n}{0} = \binom{n}{n} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{2} = \frac{n(n-1)}{2}.$$

Why divide by $k!$? Each subset of size k gives exactly $k!$ distinct lists depending on the order of its elements. The number of lists without repetition is therefore $k!\binom{n}{k}$.

Identities and proofs

$$\binom{n}{k} = \binom{n}{n-k}, \qquad \binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}, \qquad \sum_{k=0}^{n} \binom{n}{k} = 2^n.$$

Pascal's rule above is used for $0 \leq k < n$; the endpoint formulas can be read directly with the values 1.

Symmetry. Taking the complement sets up a one-to-one correspondence between the subsets of size k and those of size $n-k$.

Pascal, combinatorial proof. In a set of $n+1$ elements, fix one element a. A subset of size $k+1$ contains a (choose k other elements from n) or does not contain it (choose $k+1$ elements among n). The two cases are disjoint and cover every possibility.

Pascal, algebraic proof. For $0 < k < n$, put over a common denominator:

$$\frac{n!}{k!(n-k)!} + \frac{n!}{(k+1)!(n-k-1)!} = \frac{n!((k+1)+(n-k))}{(k+1)!(n-k)!} = \frac{(n+1)!}{(k+1)!(n-k)!}.$$

Edge cases are dealt with directly.

Number of subsets. For each of the n elements, choosing "present" or "absent" gives 2^n subsets. Classifying them by size, we count $\sum_{k=0}^{n}\binom{n}{k}$: both counts are equal.

Extension mentioned in the scheme of work. Expanding $(a+b)^n = \sum_{k=0}^{n}\binom{n}{k}a^{n-k}b^k$ is understood by choosing, in each of the n factors, the term a or b. This formula is not used as a prerequisite for the written tests in this booklet.

Define what is being counted

9.1 Order and repetition: two separate questions

First use: S22.

Worked example · Two letters chosen from A, B, C

With order and repetition, the nine lists are AA, AB, AC, BA, BB, BC, CA, CB, CC. With order without repetition, six lists remain. Without order or repetition, the three objects are $\{A, B\}$, $\{A, C\}$ and $\{B, C\}$. The result changes because the counted object changes, not because we choose another formula.

1st letter / 2nd letter	A	B	C
A	AA	AB	AC
B	BA	BB	BC
C	CA	CB	CC

Cross out the diagonal to forbid repetition; group AB and BA to ignore order.

A1 · Immediate application List the codes of two distinct digits taken from $\{0, 1, 2\}$. How many represent two-digit numbers? Explain the difference between a code and a number.

Property · Addition and product

Disjoint cases are counted by addition. Successive choices are counted by multiplication if each stage offers the stated number of possibilities, given the previous choices. For n distinct objects, the lists of length k without repetition number $n(n-1)\cdots(n-k+1)$; the number of complete permutations is $n!$.

A2 · Partial guidance An outfit has one top from four and one pair of trousers from three. One top and a pair of trousers are red. Count the outfits containing both red items, then exactly one red item.

9.2 Forgetting order without losing possibilities

First use: S30.

Definition · Binomial coefficient

$\binom{n}{k}$ is the number of subsets with k elements of a set with n elements. For $0 \leq k \leq n$, $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. In particular $\binom{n}{0} = \binom{n}{n} = 1$.

Proof. Each subset with k elements gives $k!$ distinct ordered lists. All the subsets give the same number of lists. Dividing the $n!/(n-k)!$ lists without repetition by $k!$ thus counts each subset exactly once. □

Worked example · A team and roles

A team of three students out of eight can be chosen in $\binom{8}{3} = 56$ ways. If one of the three becomes captain, there are $56 \times 3 = 168$ choices. We recover $\binom{7}{2} = 168$ by first choosing the captain. To assign three distinct roles, there would be $8 \times 7 \times 6 = 336$ possibilities.

A3 · Independent choice Ten pupils, six in Terminale and four in Première, form a team of three. Count the teams containing exactly two Terminale students, then at least one student from each level, by two methods.

A4 · Fix a count "Choosing two delegates out of five gives 5 $\times 4 = 20$ possibilities." Explain the mistake if the roles are identical. In which situation would the result 20 be correct?

9.3 Two proofs of Pascal's rule, two readings of a sum

First use: S35 and S37.

Property · Proven identities

For $1 \leq k \leq n$, $\binom{n+1}{k} = \binom{n}{k} + \binom{n}{k-1}$. In addition, $\sum_{k=0}^{n} \binom{n}{k} = 2^n$.

Proof. Single out one element in a set of size $n+1$. The subsets of size k that do not contain it are $\binom{n}{k}$; those which contain it require $k-1$ other elements, i.e. $\binom{n}{k-1}$. The two cases are disjoint and exhaustive. By calculation, put the two fractions over the same denominator $k!(n-k+1)!$ gives a numerator $n![(n-k+1)+k] = (n+1)!$. Finally, a subset of a set with n elements is built by n binary choices (take each element or not), i.e. 2^n subsets; the sorting them by size gives the sum. □

Worked example · A distinguished element

Among five pupils including Lina, a team of two contains Lina (four choices for her partner) or does not contain her ($\binom{4}{2} = 6$ choices). Hence $\binom{5}{2} = 4 + 6 = 10$.

A5 · Immediate application From row 1, 4, 6, 4, 1 of Pascal's triangle, construct the next one and check the sum of its coefficients. Interpret the coefficient 10.

9.4 Complement and double counting

First use: S35.

Worked example · At least one occurrence

A four-digit code in $\{0,1,2,3,4\}$ allows repetition. There are 5^4 codes and 4^4 codes without zero, therefore $5^4 - 4^4 = 369$ codes with at least one zero. The calculation 4×5^3 would count codes containing several zeros several times.

A6 · Choice of strategy Among these codes, count those that contain exactly two zeros. Then count those whose digit sum is odd, without listing the 625 codes.

A first step: For the second question, classify according to the number of positions holding an odd digit: one or three.

One minute, spoken aloud. Explain why systematically dividing a product by a factorial may be wrong: first state what is regarded as the same result.

Delayed review · S41 then S85. Recover the binomial coefficient as a number of paths containing exactly k successes; compare this interpretation with that of a team.

Investigation

Order and repetition in counting - 30 min.

Six different students volunteer to organise a debate.

1. Two pupils are chosen to form a pair without assigning roles. How many pairs can be formed?
2. A president and a secretary are appointed, who must be two different pupils. How many of assignments of the two roles are possible?
3. Three successive presentations are planned. The same pupil may present several times. How many schedules of three presentations are possible?
4. We start again with only four pupils, denoted by A, B, C and D. Write all pairs and all the assignments of the two roles. Then explain why there are 64 schedules of three presentations, without listing them.
5. For each of the three ways of choosing, state whether order matters and whether the same person can be chosen several times.

Progressive exercises

E1 · Direct application. A code is a sequence of four digits chosen from 0, 1, . . . , 9. The repetitions and the digit 0 in first position are allowed. How many codes contain at least one 0?

E2 · Application and justification. Ten athletes run a race with no ties. How many different podiums, with a first, second and third place, can be obtained?

E3 · Method to organise. From seven distinct people, a group of three people is formed containing a person designated in advance. How many groups can be formed?

E4 · Choice of strategy. On a grid, we start from $(0;0)$ to reach $(4;3)$ by performing exactly four moves of one unit to the right and three upwards. How many different paths can be followed?

E5 · Synthesis or proof. A class has eight girls and seven boys. How many groups of four students containing exactly two girls can be formed?

Work from the textbook or a past paper

Barbazo, p. 294, exercise 37. At home after S35: deal with both questions. State whether order matters and whether a card can be chosen more than once before using a binomial coefficient.

Homework

DM9 - Forming a committee - 45 min, after S35.

A team consists of five teachers and seven students, all distinct. A committee is formed of four people.

1. How many different committees can be formed?
2. How many committees contain exactly two pupils?

3. How many committees contain at least one teacher and at least one pupil?

4. From the members of a committee in question 2, we appoint a teacher as chair and a student secretary. Count the possible choices of committee together with these two roles.

5. By counting the committees according to their number of teachers, prove that

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4}.$$

MCQ A · order and repetition

After S25; MCQ B after S35.

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

1. A three-digit code chosen from 0, 1, 2, 3, with repetition allowed, gives:

A. 4×3 B. 3^4

C. 4^3 D. 4!

2. Two distinct roles are given to two people among five. The number of assignments is:

A. 10 B. 20

C. 25 D. 120

3. The number of permutations of four distinct objects is:

A. 16 B. 12

C. 4 D. 24

4. For two disjoint finite sets A, B, $|A \cup B|$ is:

A. $|A| + |B|$ B. $|A||B|$

C. $|A| - |B|$ D. $|A \cap B|$

5. With two cards A and B, the lists of length 2 with repetition are:

A. AB, BA B. AA, BB

C. AA, AB, BA, BB D. A, B

MCQ B · complement and double counting

For each question, only one option is correct. Justify your choice orally before looking at the solutions.

6. Choosing three pupils among eight with no assigned role is counted by:

A. 8^3 B. $\binom{8}{3}$

C. $8 \times 7 \times 6$ D. 3^8

7. The number of subsets of a five-element set is:

A. 5! B. 10

C. 25 D. 32

8. Pascal's rule gives $\binom{6}{3}$ equal to:

A. $\binom{5}{2}+\binom{5}{3}$ B. $\binom{5}{2}\binom{5}{3}$

C. $2\binom{6}{2}$ D. $\binom{6}{4}+1$

9. An urn contains 3 red and 2 blue distinct balls. The number of pairs without replacement containing at least one red ball is:

A. 6 B. 10

C. 9 D. 3

10. Six people are split into two numbered groups of three. The number of ways is:

A. 10 B. 20

C. 40 D. 6!

True or false · justification expected. For each statement, indicate whether it is true or false, then write a proof or a counterexample.

1. $\binom{n}{0}=0$.

2. Counting with the complement can avoid distinguishing several cases.

3. Drawing three objects simultaneously always gives the objects an order.

MCQ solutions

1. **C.** Three independent positions with four choices each, including zero at the start: $4^3=64$.
2. **B.** $5\times 4=20$. The order of the roles matters and the same person cannot hold both.
3. **D.** $4!=4\times 3\times 2\times 1=24$.
4. **A.** Each element is counted once. Without disjointness, the intersection would have to be subtracted.
5. **C.** A list keeps the order and allows a card to be reused; there are $2^2=4$ outcomes.
6. **B.** The group is not ordered, so $\binom{8}{3}=56$; the ordered product counts each group $3!$ times.
7. **D.** For each element, two decisions: included or not. We get $2^5=32$, including the empty subset.
8. **A.** Separate the groups containing a fixed element (choose two from five) from those not containing it (three from five).
9. **C.** $\binom{5}{2}-\binom{2}{2}=10-1=9$. Another calculation: $3\times 2+\binom{3}{2}=9$. Six counts only the mixed pairs.
10. **B.** Choosing the three members of group 1 determines group 2: $\binom{6}{3}=20$. Dividing by 2 would be correct only for unnumbered groups.

True/false. False: there is an empty subset, so $\binom{n}{0}=1$. True: question 9. False: a simultaneous draw gives a subset; adding an order changes the object being counted.

Solutions to the applications and guidance

Application 1. Codes: 01,02,10,12,20,21. Only the last four represent numbers with two digits. A code may start with zero.

Application 2. Two red items: one outfit. Exactly one: $1 \times 2 + 3 \times 1 = 5$ outfits. The two cases are disjoint.

Application 3. Exactly two Terminale students: $\binom{6}{2}\binom{4}{1} = 60$. Mixed group: $\binom{10}{3} - \binom{6}{3} - \binom{4}{3} = 96$; or $\binom{6}{2}\binom{4}{1} + \binom{6}{1}\binom{4}{2} = 60 + 36 = 96$.

Application 4. Each unordered pair is counted twice; there are $\binom{5}{2} = 10$ pairs. For a main member and a substitute, the roles distinguish the choices, hence 20.

Application 5. Row: 1, 5, 10, 10, 5, 1, sum 32. The first 10 counts the subsets of size 2 of a set of size 5; the second those of size 3.

Application 6. Two zeros: $\binom{4}{2}4^2 = 96$. Of five digits, three are even and two odd. Sum odd: $\binom{4}{1}2\,3^3 + \binom{4}{3}2^3\,3 = 216 + 96 = 312$.

Consolidation

Ask before each calculation: "What is a possible result? When are two results different?" Allow lists and trees in the small case, then have students make explicit what generalises. The parity calculation can remain a supported challenge; the case-by-case method is enough.

Teaching notes and solutions

Guidance for the teacher

The common test S28 comes before the combinations in S30: include only principles and ordered lists. Written test 3 may use combinations, but not Pascal's rule taught in S35. Have students state what is being counted before any formula. For struggling students, use cards with names; does a permutation of cards change the object? The binomial theorem, mentioned in the progression, is an extension: keep it explicitly optional.

Solutions to the activity

1. A pair is a subset of two pupils among six: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

2. The president can be chosen in six ways, then the secretary from among the other five students: $6 \times 5 = 30$ assignments.

3. Each task may be given to any of the six pupils, including a pupil already chosen. The three successive choices give $6^3 = 216$ sequences.

4. With four cards, the pairs are AB, AC, AD, BC, BD, CD : six choices. The twelve pairs of roles are:

$$\begin{array}{lll}(A,B) & (A,C) & (A,D)\\ (B,A) & (B,C) & (B,D)\\ (C,A) & (C,B) & (C,D)\\ (D,A) & (D,B) & (D,C)\end{array}$$

   The number of sequences of three interventions is $4^3 = 64$, since each position has four choices.

5. For a pair, order does not matter and no pupil is repeated. For roles, order matters, without repetition. For the interventions, order matters and repetition is allowed.

E1. There are 10^4 codes in total. Those containing no 0 have nine choices for each of the four positions, i.e. 9^4 codes. Using the complement, the required number is $10^4 - 9^4 = 3439$.

E2. There are ten choices for gold, then nine for silver and eight for bronze. The multiplication principle gives $10 \times 9 \times 8 = 720$ podiums. Changing the order changes the podium.

E3. The required person is already chosen. It remains to choose a subset of two people among the six others: $\binom{6}{2} = \frac{6\times5}{2} = 15$.

E4. A path is determined by the positions of its four steps to the right among seven positions. The three other steps are then forced. We therefore count $\binom{7}{4} = 35$ paths.

E5. Choose two girls out of eight, then two boys out of seven. Each pair of choices gives exactly one group: $\binom{8}{2}\binom{7}{2} = 28 \times 21 = 588$.

Solutions to homework DM9

1. Order does not matter: there are $\binom{12}{4} = \frac{12\times11\times10\times9}{24} = 495$ committees.

2. Two pupils are chosen from seven and two teachers from five: $\binom{7}{2}\binom{5}{2} = 21 \times 10 = 210$ committees.

3. From the 495 committees we remove the $\binom{5}{4} = 5$ committees made only of teachers and the $\binom{7}{4} = 35$ committees made up only of students. These categories are disjoint: $495 - 5 - 35 = 455$.

4. Each of the 210 committees has two choices of teacher president and two choices of student secretary: $210 \times 2 \times 2 = 840$ arrangements.

5. Let us classify the committees by the number k of teachers, with $0 \leq k \leq 4$. The category k contains $\binom{5}{k}\binom{7}{4-k}$ committees. The five categories are disjoint and cover all the committees. By addition,

$$\sum_{k=0}^{4} \binom{5}{k}\binom{7}{4-k} = \binom{12}{4} = 495.$$

Solutions to the textbook or past-paper work

A hand contains five distinct cards; order does not matter. There are therefore $\binom{52}{5} = 2{,}598{,}960$ hands. For a hand containing the ace of spades, this card is already chosen; the other four are chosen from the remaining 51 cards. There are $\binom{51}{4} = 249{,}900$ hands. This result is not multiplied by 5: the hand has no first or last position.